

Gem cuts

Stones are reshaped to enhance their beauty and to increase their value. A finished gem can be many times the value of its rough, and it is also far more sellable. Gemstone rough may be sawn to remove poor areas, to separate valuable areas from within a larger stone, or to create a preliminary shape. Arriving at the final shape and form of a gem (its “cut”) then involves various stages of grinding and polishing. The cut used on a particular piece of gem rough is determined by a number of factors in combination: the shape of the rough, the position

of its flaws, its cleavage, the best orientation to display its colour (in the case of star-stones, for example, the cut is oriented so that the star is centred in the finished stone), and the most suitable cabochon shape if the stone is translucent or opaque.

Polished stones

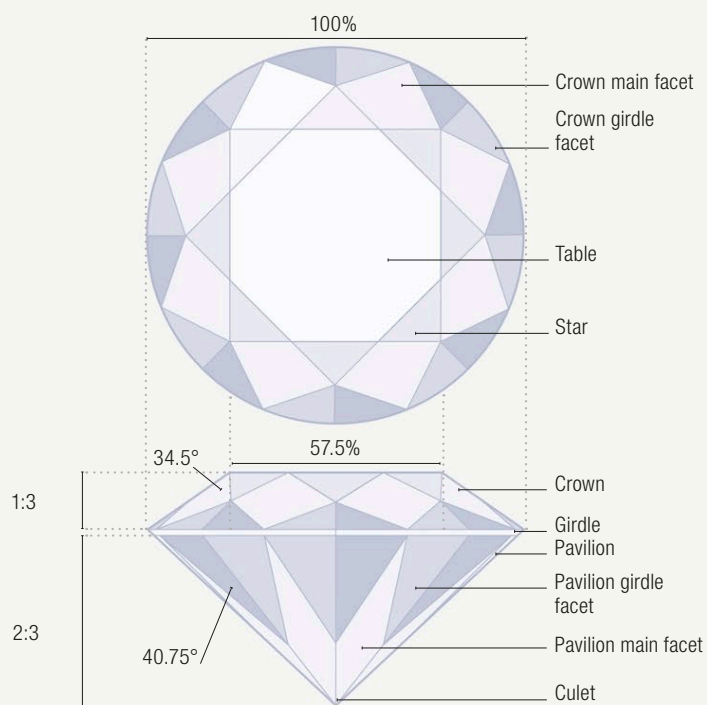
If a stone is described as “polished”, it can be anything from a slice of opaque gem material polished on the flat sides for use as a pendant to the intricate and detailed carving of a fine cameo.

The parts of a gem

Jewel terminology

Whatever the gem cut being used, there are a few fundamental parts: the crown, pavilion, and table facet. The other facets and the proportion between the crown and pavilion affect a gem’s brilliance. Whether on a round, brilliant, or rectangular emerald cut, the facets are given the same names and fill the same relative positions on the gem. The

angles at which the facets are cut are determined by the refractive index, and the cutter uses a set of tables to find the suitable face angles for each type of material. The angles shown here are for diamond. The usual ratio of the crown to the pavilion is 1:3, but this can vary depending on the angle of the crown mains.

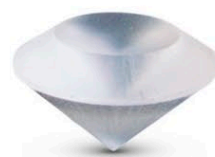


Cutting techniques

The basic stages of sawing, grinding, and polishing are common to all lapidary processes, but each of the three requires skills and tools unique to that particular step. It is not unusual for a gem cutter to be able to do all three.



Choice of rough A gem rough is chosen for its colour, size, clarity, shape, and freedom from flaws, fractures, and inclusions. If imperfections are present, the cut is oriented in such a way as to conceal them as much as possible.



Choice of cut The cut is chosen based on the shape of the rough, and the desired colour and brilliance of the finished stone. The rough is sawn to the general shape of the stone or to provide the table facet on the top half of the stone, called the crown.



Faceting begins The best brilliance is achieved when facets are angled and positioned correctly. Large facets are cut first. On a brilliant cut there are eight on the top and eight on the bottom. Usually the pavilion is faceted first, then the crown.



Further faceting Further smaller facets are added around the larger facets on the crown at the top and the pavilion at the bottom. On a brilliant cut there are 40 of these smaller ones. Each facet is cut to maximize its optical performance.



Finished off After all the final facets have been cut, the gem is then polished to remove any cutting scratches and to improve its lustre. This stage can take place alongside the faceting process – as preferred by most cutters – or afterwards.